



A progressive framework for tooth and substructure segmentation from cone-beam CT images

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ARTICLE INFO

Keywords:

Tooth and substructure segmentation
CBCT images
Hybrid features
Center clustering
Skeleton

ABSTRACT

Background: Accurate segmentation of individual tooth and their substructures including enamel, pulp, and dentin from cone-beam computed tomography (CBCT) images is essential for dental diagnosis and treatment planning in digital dentistry. Existing methods for tooth segmentation based on CBCT images have achieved substantial progress; however, techniques for further segmentation into substructures are yet to be developed. **Purpose:** We aim to propose a novel three-stage progressive deep-learning-based framework for automatically segmenting 3D tooth from CBCT images, focusing on finer substructures, i.e., enamel, pulp, and dentin.

Methods: In this paper, we first detect each tooth using its centroid by a clustering scheme, which efficiently determines each tooth detection by applying learned displacement vectors from the foreground tooth region. Next, guided by the detected centroid, each tooth proposal, combined with the corresponding tooth map, is processed through our tooth segmentation network. We also present an attention-based hybrid feature fusion mechanism, which provides intricate details of the tooth boundary while maintaining the global tooth shape, thereby enhancing the segmentation process. Additionally, we utilize the skeleton of the tooth as a guide for subsequent substructure segmentation.

Results: Our algorithm is extensively evaluated on a collected dataset of 314 patients, and the extensive comparison and ablation studies demonstrate superior segmentation results of our approach.

Conclusions: Our proposed method can automatically segment tooth and finer substructures from CBCT images, underlining its potential applicability for clinical diagnosis and surgical treatment.

1. Introduction

In modern dentistry, the accurate segmentation of individual tooth and their substructures (i.e., enamel, pulp, and dentin) from cone-beam computed tomography (CBCT) [1] images is pivotal. Tooth models can provide a complete three-dimensional tooth to assist in orthodontic and implant planning, while the segmentation of tooth substructures offers even greater potential for aiding tooth diagnosis (e.g., caries, pulpitis, and apical lesions) and treatment planning (e.g., crown restoration planning, and root canal therapy), as well as age analysis in forensic science. Considering that manual annotation on CBCT images is extremely time-consuming and requires specialized knowledge, the development of an efficient and robust algorithm for the automatic extraction of 3D teeth and substructures models from CBCT images has become a focus of widespread attention in the research community.

However, the precise segmentation of individual tooth and substructures poses challenges due to the following reasons. Firstly, metal

artifacts induced by metal braces or calcification increase the difficulty of automatic tooth segmentation and identification (Fig. 1(a)). Secondly, the tooth boundary in CBCT images is often indistinct, particularly the boundary between tooth roots and surrounding alveolar bones, as well as the common boundary of touching teeth (i.e., lower and upper teeth in the natural bite condition) (Fig. 1(c)). From the substructure perspective, the internal boundaries of tooth substructures exhibit high similarities, such as low contrast at the interface between dentin and pulp (Fig. 1(c)). Moreover, the diversity and delicate shapes of dental pulp across different teeth affect the accuracy of pulp segmentation (Fig. 1(d)).

To address these challenges, conventional methods [2–7] relying on handcrafted features, including image intensity and well-defined shape priors, have been explored to segment teeth objects from CBCT images. Unfortunately, considering the limitations of low-level feature representations in contextual capability and robustness, these methods often

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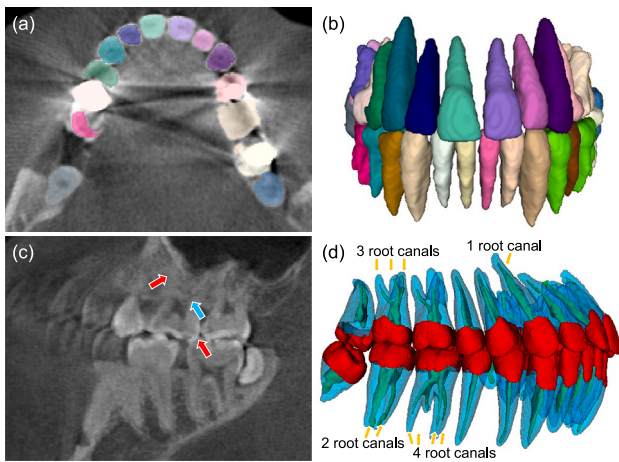


Fig. 1. Overview of teeth and their structures, along with challenges in segmentation. (a) and (b) display the tooth segmentation results of a CBCT image with metal artifacts; (c) two red arrows point out ambiguous boundaries with alveolar bone and adjacent tooth, and also a blue arrow shows low contrast between pulp and dentin; (d) illustrates the segmentation of tooth substructures, where the red, green, and blue parts represent enamel, pulp, and dentin, respectively. Note that incisors, canines, and molars exhibit a variable number of root canals.

lead to inaccurate segmentation. This is especially true in CBCT images taken before dental treatment, where extreme cases such as tooth misalignment and decay problems are common. In addition, most of these methods need extra user annotations or specially designed teeth templates, reducing efficiency. Recently, convolutional neural networks (CNNs) have achieved remarkable progress in automatic tooth segmentation. In the first stream, some works [8,9] extend the 2D instance segmentation framework, Mask R-CNN [10], to a 3D version. Although these methods have achieved promising performance, they cost enormous computational resources since the region proposal network based on pre-defined anchors generates plenty of redundant proposals in the detection stage. Additionally, another approach [11] relies on tooth contour information to separate and segment individual tooth. Yet, these methods are dependent on contour extraction results, often producing blurred boundaries near thin artifacts.

Currently, only a few tasks perform detailed substructure segmentation on teeth. Existing models like V-Net have been used to segment tooth enamel, dentin, pulp, and filling on the occlusal slice, providing benchmark experiments for this underexplored area [12]. However, these are only 2D attempts and offer limited guidance. Some root canal segmentation methods in the substructure have been explored, including 2D methods [13–15] and 3D-based methods [16–18], but none have successfully segmented complete tooth substructures on CBCT images.

Considering the impact of artifacts in tooth segmentation and the lack of complete segmentation of tooth substructures, we develop a novel deep-learning-based three-stage framework for automatically segmenting teeth and corresponding substructures. In the first stage, we utilize a detection network to efficiently identify all teeth, employing clustering to find tooth centroids. Then, we design a novel attention-based hybrid feature fusion mechanism for tooth segmentation. Compared with a conventional segmentation network with a skip connection, the attention-based structure dynamically integrates the low- and high-level features that preserve the tooth shape and boundary information. This improves overall performance, especially in the challenging cases with metal artifacts and calcification. In the final stage, we design a network specifically to decompose each tooth into its enamel, pulp, and dentin structures, aligning the geometric shape of the dental pulp with the tooth skeleton. By employing the tooth skeleton as a topological guidance, we can accurately delineate the dental pulp connectivity.

Our main contributions are summarized as follows:

1. We design a novel three-stage architecture, capable of simultaneously detecting and segmenting the individual tooth and corresponding substructures with accuracy and efficiency.
2. We present a hybrid feature fusion mechanism in tooth segmentation that fortifies our model against artifacts in CBCT images. We also propose using tooth skeleton to refine pulp segmentation through topological guidance.
3. Extensive experiments using dataset collected from dental clinics demonstrate that our approach outperforms state-of-the-art methods in both accuracy and efficiency, with the ability to handle more challenging real-world clinical cases.

2. Related works

In this section, we first conduct an extensive review of the literature on tooth segmentation in CBCT images, encompassing both conventional methodologies and deep-learning-based approaches. We further delve into recent works on the segmentation of tooth substructures, with particular emphasis on pulp segmentation.

2.1. Tooth segmentation from CBCT images

Tooth segmentation in CBCT images is an essential step for orthodontic treatment. Many automatic and semi-automatic methods have been developed, demonstrating considerable success. Prior works [2–7] have investigated the way to design effective hand-crafted features to separate the tooth object from surrounding tissues, utilizing methods like graph cut-based methods, level-set-based methods, and template-based methods. However, these methods exhibit certain limitations, such as being sensitive to the definition of statistical shape priors and the variation of foreground intensities, falling short with CBCT images typically scanned in a closed bite condition of the lower and upper jaws, not adapting well to patient-specific variations in shape, respectively.

In recent years, the application of deep learning techniques to automate tooth segmentation in CBCT images has flourished [8,9,19–22]. Cui et al. [8] extend the 2D object detection and segmentation framework, Mask R-CNN [10], to a 3D variant, while Chung et al. [9] propose a pose regression neural network to align the jaws and apply a faster region-based convolutional neural network [23] to perform tooth segmentation. Although these methods have achieved leading performance, they necessitate substantial GPU memory for 3D volumetric data. Chen et al. [19] employ the multi-task Fully Convolutional Network (FCN) to simultaneously predict the semantic tooth label and boundary, which are fed into the watershed transform to separate and generate instance labels. However, this method may create artifacts, particularly in regions of close contact between teeth. Wu et al. [20] propose a center-sensitive and boundary-aware tooth instance segmentation framework. Cui et al. [21,22] present a learning-based framework that explicitly leverages the comprehensive geometric guidance provided by a hierarchical tooth morphological representation for tooth instance segmentation. Inspired by these articles, we design our tooth detection and segmentation method. First, we utilize the centroid voting scheme to represent teeth robustly. Second, our attention-based feature fusion scheme greatly improves our anti-artifact ability in our tooth segmentation.

2.2. Tooth substructure segmentation

Research on finer substructure segmentation is relatively scarce. L. Schneider et al. [12], established a benchmark for tooth structure segmentation (e.g., enamel, dentin, pulp, fillings, crowns) using well-known deep learning models like V-Net, U-Net, and ResNet. However, the provided dataset comprises 2D dental occlusal radiographs, making it challenging to accurately determine the three-dimensional shape through overlapping structures.

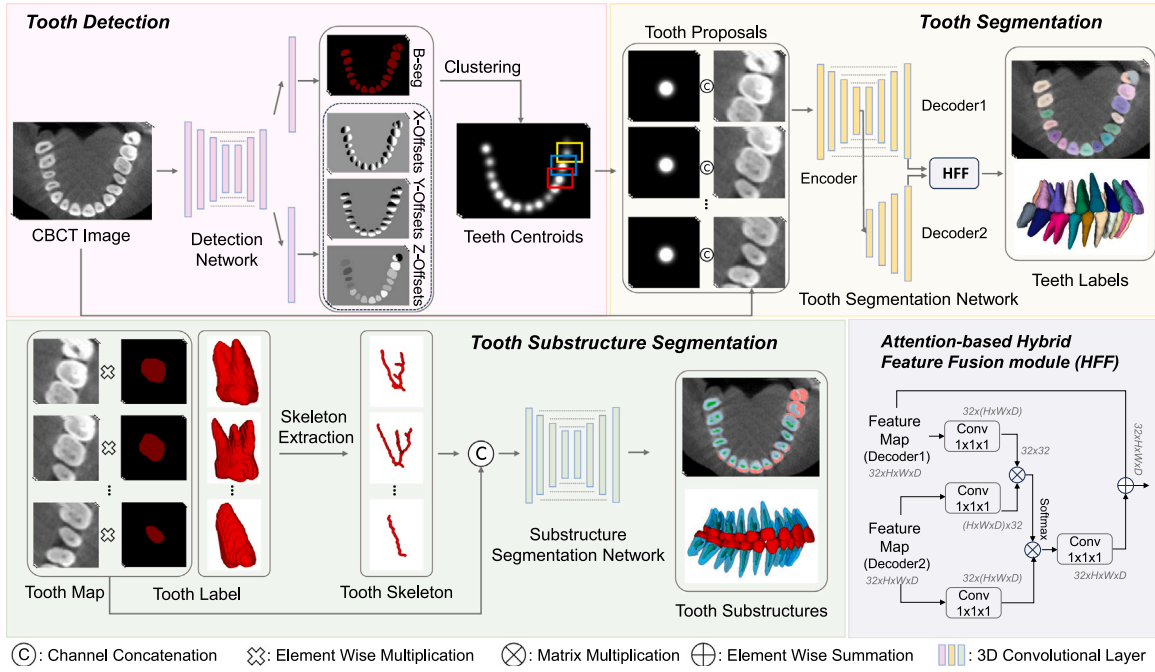


Fig. 2. Overview of our tooth and substructure segmentation framework. Three principal components are included: tooth detection, tooth segmentation, and tooth substructure segmentation.

Several substantial research has been conducted on pulp segmentation, specifically, using traditional threshold-based or optimization-based methods for tooth root canal segmentation in CBCT images [13–15]. For instance, J Michetti et al. [13] evaluate the impact of image settings on endodontic segmentation, while Wang et al. [14] propose a pulp cavity region segmentation method based on plane intercept histogram of reciprocal cross-entropy. Since 2D segmentation methods may neglect spatial correlations between cross-sections, leading to discontinuous 3D reconstructions, the pursuit of 3D strategies is of high significance. Duan et al. [17] propose a two-stage deep-learning solution for accurate tooth and pulp cavity segmentation, which extracts bounding boxes for individual teeth and then achieves tooth and pulp segmentation of single-rooted and multi-rooted teeth. Zheng et al. [16] train a deep learning model for coarse segmentation of 3D pulp cavities in the first molar and refined the coarse segmentation using a level set model for volume quantification and age estimation. Wang et al. [18] propose DentalNet and PulpNet for tooth instance segmentation and root canal segmentation from CBCT images, respectively. However, existing methods target the pulp cavity, overlooking other vital substructures like enamel and dentin. In this work, we propose a novel framework for automatic tooth and their substructure segmentation from CBCT images, particularly enhancing pulp segmentation due to its complex, elongated anatomical structure.

3. Methods

An overview of our framework is shown in Fig. 2, comprised of three stages: tooth detection, tooth segmentation, and finer substructure segmentation.

3.1. Tooth detection by centroid clustering

In our framework, the presence of individual teeth is signified using an intrinsic property, i.e., tooth centroid. The centroid point, in contrast to other feature points on a tooth, consistently lies within the tooth's form, independent of its category, shape, or location. Consequently, tooth detection is inherently translated to tooth centroid regression, with a proposed clustering mechanism to effectuate this regression.

3.1.1. 3D offset and binary maps regression

Directly regressing teeth centroids proves infeasible for certain teeth, such as the incisors on the low jaw, which often possess slender crowns and are tightly packed with adjacent teeth. To address this challenge, we introduce the binary segmentation map and the offset map. Specifically, given the input CBCT images, the binary segmentation map indicates that each voxel in the input is either part of the foreground tooth object (value equals 1) or the background (value equals 0), while the offset map delineates the spatial offset vector of each foreground voxel, pointing to its corresponding centroid.

3.1.2. Teeth centroid clustering

To derive the tooth centroid from the regressed binary segmentation and offset maps, we adopt a robust clustering method [24] that efficiently identifies the density peaks as cluster centers from a density map P . Initially, we explain the calculation of the density map for the clustering algorithm. Formally, given the offset map O and binary segmentation map B , we obtain the teeth centroids density map P as follows. For any voxel i , the 3D coordinates are represented by $C_i = (x_i, y_i, z_i)$, and for any foreground voxel j , i.e., $j : B_j = 1$, we add its offset vector $O_j = (\Delta x_j, \Delta y_j, \Delta z_j)$ with its 3D coordinate C_j to get the new 3D coordinate C_k after rounding the values to integers, i.e., $C_k = \text{Round}(O_j + C_j)$. Then the density map P is calculated as:

$$P_k = \mathbb{I}(C_k), \quad (1)$$

where the indicator function counts the frequency of the 3D coordinate C_k being pointed by the foreground voxels. Note that, P_k has the scalar value indicating how salient voxel k is regarded as a tooth centroid. Other than the density map, we also define a distance map D_i in the Euclidean space for any voxel i conditioned on the density map:

$$D_i = \min_{j: P_j < P_i} \|C_i - C_j\|_2. \quad (2)$$

For the voxel with the highest density, we calculate the $D_i = \max_j \|C_i - C_j\|_2$.

Finally, the clustering algorithm aims to find the set of cluster centers (i.e., tooth centroids) T_i with larger density P_i and relatively bigger neighboring distance D_i simultaneously, defined as:

$$T_i = (P_i > \alpha) \cap (D_i > \beta), \quad (3)$$

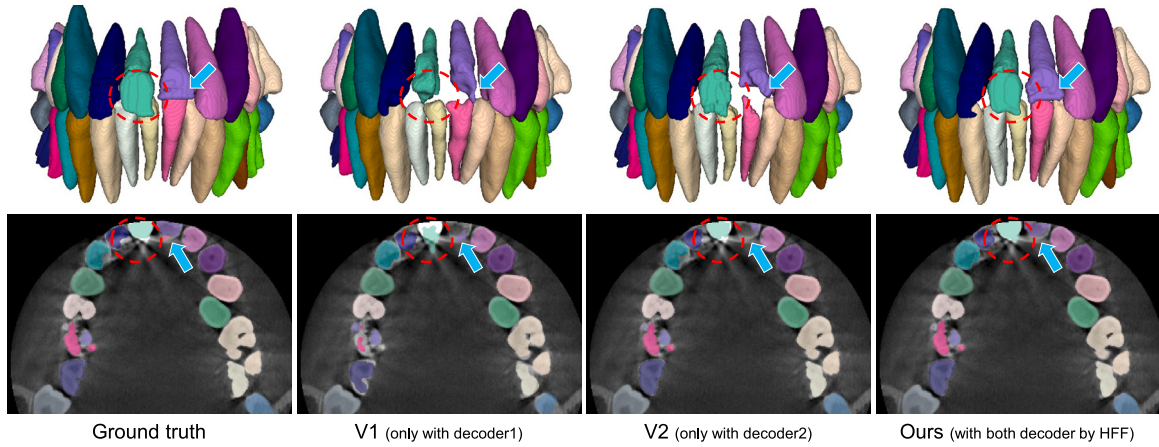


Fig. 3. Qualitative comparison of the segmentation results on the CBCT images with metal artifacts. The first row presents the 3D reconstructed results while the second row shows the corresponding 2D results in a zoom-in view of a specific slice. Exemplary areas are highlighted by blue arrows and red circles.

where α and β refer to the density threshold and neighboring distance threshold, respectively. We use $\alpha = 30$ and $\beta = 5$ in all the experiments.

3.1.3. Network losses

To train the tooth detection subnetwork, we deploy two losses for supervision. Specifically, we combine the cross entropy loss with the dice loss to supervise the binary segmentation that separates the tooth object from the background tissues (denoted as L_{seg}^b). Concurrently, for the supervision of the offset map, we utilize the smooth L_1 loss on the voxels that belong to the tooth object (denoted as $L_{reg}^{smoothL_1}$). Thus, the total detection loss L_{td} is the summation of the segmentation and regression losses, defined as:

$$L_{td} = L_{seg}^b + L_{reg}^{smoothL_1}. \quad (4)$$

3.2. Tooth segmentation via fused hybrid features

Guided by the regressed teeth centroids, we initially generate the tooth proposal for each individual tooth, which will be subsequently input into the attention-based feature fusion module to produce the hybrid feature for tooth segmentation.

3.2.1. Tooth proposal generation

The tooth proposal for each individual tooth includes a tooth map and a Gaussian-like guidance map. Specifically, centered at each predicted tooth's centroid $T_k, k = 1, \dots, K$, we directly crop the input CBCT image into the tooth map $M(k)$. Then, we calculate a guidance map $\Phi(k)$ on the predicted teeth's centroids using a Gaussian function with a small fixed standard deviation $\delta = 5$, as defined as:

$$\phi_u(k) = \exp\left(-\frac{\|u - T(k)\|^2}{2\delta^2}\right), \quad (5)$$

where u refers to the spatial coordinate of the guidance map $\Phi(k)$. Finally, we utilize a channel concatenation from $M(k)$ and $\Phi(k)$ and set it as the k th tooth proposal.

3.2.2. Hybrid feature fusion (HFF)

Upon generating teeth proposals, our objective is to isolate the specific instance of a tooth within each proposal. The V-Net, acclaimed for its promising performance in myriad medical image analysis applications, is thus adopted as the backbone of the segmentation network. Nevertheless, we observe that the segmentation performance is sensitive to metal artifacts in CBCT images, as shown in Fig. 3. The V1 is the baseline which uses the skip connection between the encoder and decoder1, while the V2 decodes solely from the high-level feature of the encoder output by decoder2. As deduced from Fig. 3, such artifacts are predominantly encoded within the low-level features of

the convolutional layers in the encoder and transferred to the decoder via the skip link. Moreover, V2, in comparison to V1, better preserves the 3D tooth shape information in areas with metal artifacts, but at the expense of losing intricate details, especially at the tooth boundary without direct access to low-level features.

Motivated by these insights, we propose a novel segmentation framework that capitalizes on hybrid features. Specifically, we augment the encoder output with an additional branch equipped with decoder2. To synergistically leverage both the low-level image feature and the high-level shape characteristics, we employ an attention module [25] adeptly balances the weight of the two features in conjunction. The fused feature, as well as the features from the decoder1 and decoder2 branches, are channeled through several convolutional layers to derive the final segmentation results.

Fig. 2 illustrates the architecture of the hybrid fusion module, commencing with the transmission of feature maps from both decoder1 and decoder2 through convolutional layers. Subsequently, matrix multiplication is executed, succeeded by a softmax layer to obtain the channel-wise attention map with a size of 32×32 . Further matrix multiplication is performed between this attention map and the feature map of decoder2, succeeded by another convolutional layer. Ultimately, the product of this final multiplication is amalgamated with the V1 feature map, forming the definitive feature employed to predict tooth labels.

3.2.3. Loss function of tooth segmentation

To stabilize network training and optimize performance, we employ a combined approach for supervising the segmentation tasks, using both cross-entropy loss and dice loss. This combination is formally expressed as:

$$L_{seg} = \lambda(L_{dice} + L_{ce}), \quad (6)$$

where λ is the balancing weight and is empirically set as 0.5 in our experiments.

3.3. Tooth substructure segmentation

Subsequent to the tooth segmentation phase, we ascertain the tooth label consisting of the enamel, pulp, and dentin. To attain labels of individual substructures, we feed the tooth map and its corresponding tooth label into the substructure segmentation network. Notably, as tooth segmentation offers more dependable delineation of entire tooth boundaries, we utilize the predicted tooth label to extract the foreground from the tooth map directly. This focuses the segmentation process on the substructure segmentation within each tooth.

Table 1
Statistical comparison with the heatmap regression method on our evaluation metrics.

Methods	Localization (mm)	Dice (%)	Precision (%)	95% HD (mm)	ASD (mm)
HM_{reg}	1.06 ± 0.30	94.69 ± 0.12	96.78 ± 0.07	0.455 ± 0.87	0.077 ± 0.30
HFF-Net(Ours)	0.71 ± 0.46	95.78 ± 0.06	96.86 ± 0.07	0.449 ± 0.86	0.075 ± 0.29

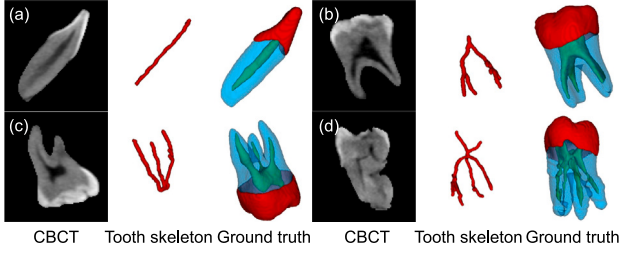


Fig. 4. Typical cases for tooth skeletons of incisors and molars. Notice that all tooth skeletons share similar shapes with their corresponding pulps.

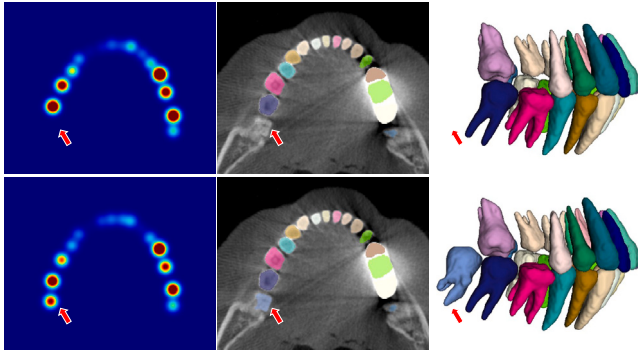


Fig. 5. Visual comparison of different centroid detection schemes, based on the heatmap regression (the 1st row) and offset voting (the 2nd row). The first column represents the centroid voting map, while the subsequent columns indicate the corresponding segmentation results in 2D and 3D formats.

3.3.1. Tooth skeleton

While the segmentation network reliably segments enamel and dentin, it encounters difficulties with pulp segmentation due to the two-fold challenges: (1) The pulp's morphology significantly varies across different types of teeth, and (2) The delicate and tubular structure of the pulp presents complications in preserving its continuity throughout the segmentation process. To address these problems, we propose a method that leverages the tooth skeleton, derived from the tooth segmentation outcomes, for the subsequent pulp segmentation. Specifically, we use the traditional method of Lee [26], iteratively extracting the axis of the tooth, to obtain the tooth skeleton. Considering that the skeleton has a width of only one pixel and is prone to breakage, we further apply binary image dilation to obtain a continuous and robust tooth skeleton. The tooth skeleton obtained through this method can accurately represent tooth morphology. As shown in Fig. 4, the tooth skeleton and the pulp exhibit remarkable shares, particularly in the quantity and pattern of root canals. This similarity between the tooth skeleton and the pulp is advantageous in supplying efficacious topological information that can direct pulp segmentation and maintain its structural integrity.

3.3.2. Loss function of substructure segmentation

As for the stage of tooth substructure segmentation, we design a combined dice loss function to supervise different structures (i.e., enamel, pulp, and dentin):

$$L_{sub} = \frac{1}{3}(L_{enamel} + L_{pulp} + L_{dentin}). \quad (7)$$

4. Experiments

4.1. Dataset and implementation details

We collected 314 3D CBCT scans with resolutions ranging from 0.2 to 0.4 mm, obtained from real-world clinic settings. We divided the data randomly into subsets of 70% for training, 10% for validation, and 20% for testing. Note that incomplete teeth, including wisdom teeth, and implants, are excluded in the third stage of tooth substructure segmentation.

In striking a balance between computational efficiency and segmentation accuracy, we pre-process each CBCT scan with the following step. Initially, the intensity values were clipped to the range [0, 3000], followed by normalization to the interval [0, 1]. During tooth detection, all CBCT images were normalized to $0.4 \times 0.4 \times 0.4 \text{ mm}^3$, cropped, and resized to the size of $256 \times 256 \times 256$. Moreover, the tooth patches in the second segmentation stage are cropped with the size of $96 \times 96 \times 144$ based on the tooth centroids.

4.2. Evaluation metrics

To evaluate the performance of our framework, we employ the tooth centroid localization metric to measure the detection accuracy, along with four additional metrics: Dice coefficient, Precision, 95% Hausdorff distance, and Average symmetric distance, to measure the tooth segmentation accuracy. Specifically, the tooth localization metric was computed as the distance between the predicted and the actual tooth centroid based on the ground truth segmentation label:

$$Localization = \|\Delta x + \Delta y + \Delta z\|_2, \quad (8)$$

where Δx , Δy , and Δz are the absolute difference along X , Y and Z axes, respectively. A smaller localization distance indicates higher tooth detection accuracy. The Dice metric measures the ratio of the segmentation overlap at the voxel level, which is calculated as:

$$Dice = \frac{2 \times |A \cap B|}{|A| + |B|} \times 100\%, \quad (9)$$

where A is the ground truth foreground voxels and B is the predicted foreground voxel set. Similarly, the precision metric is defined by $\frac{TP}{TP+FP}$, and TP , FN , and FP are the number of true positive, false negative, and false positive voxels. Higher Dice and precision values imply improved segmentation accuracy. The remaining metrics, 95% Hausdorff surface distance (95% HD), and Average symmetric distance (ASD) are computed on the ground truth and the segmented tooth surfaces. A smaller distance in these metrics signifies superior segmentation accuracy.

4.3. Performance of proposed method

4.3.1. Tooth detection

Our voting-based tooth proposal generation component in the tooth detection stage achieves excellent centroid localization performance for subsequent segmentation. For tooth centroid detection in the first stage, there are different choices. One alternative design is to directly regress a heatmap, where the center indicates the tooth centroid. For qualitative and quantitative evaluation, we compare our proposed method with the heatmap regression approach (denoted HM_{reg}), wherein we replace the offset regression and voting steps while maintaining the other phases. Intuitively, the direct regression of the centroid from the heatmap may struggle to generate accurate center positions (see

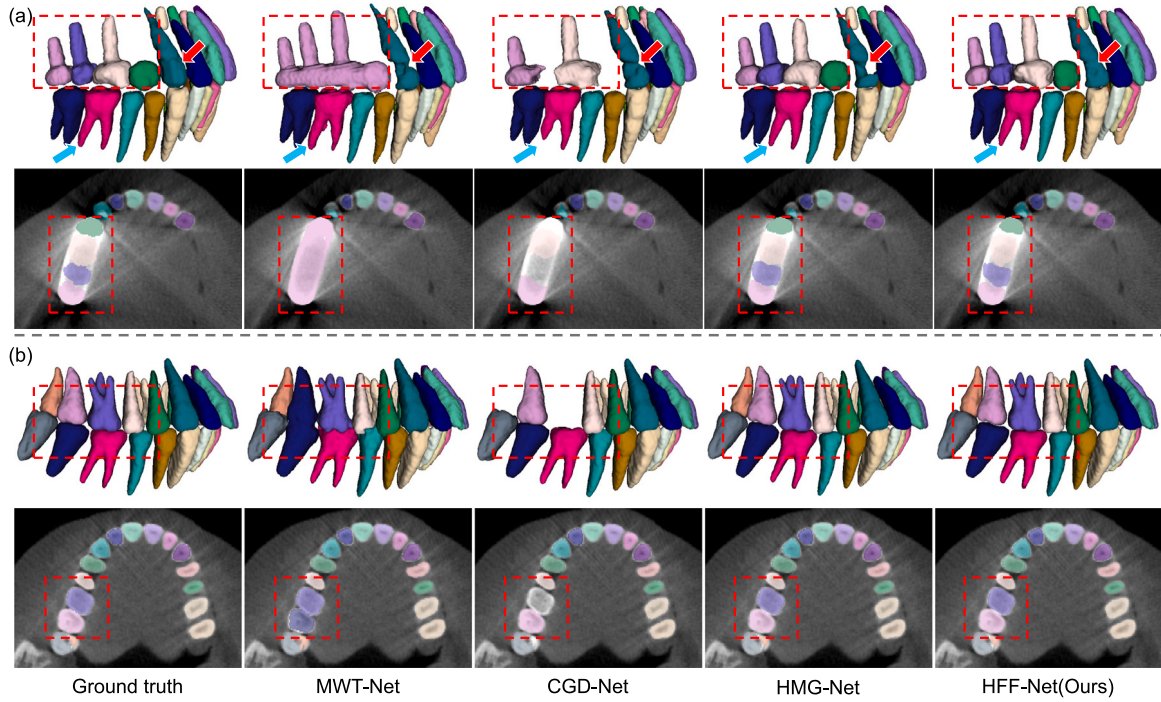


Fig. 6. Qualitative comparison with state-of-the-art methods on two typical examples. The whole tooth shape reconstructed from the segmentation and the slice-based segmentation labels are shown in the first and second rows, respectively.

Table 2

Statistical results of our ablation study by comparing with alternative network structures in terms of tooth segmentation accuracy.

Methods	Dice (%)	Precision (%)	95% HD (mm)	ASD (mm)
V1	90.65 ± 0.16	93.58 ± 0.09	0.604 ± 1.03	0.144 ± 0.36
V2	94.46 ± 0.12	95.60 ± 0.07	0.469 ± 0.94	0.082 ± 0.32
HFF-Net (Ours)	95.78 ± 0.06	96.86 ± 0.07	0.449 ± 0.86	0.075 ± 0.29

the 1st row of Fig. 5), leading to significant miss-detection errors. Fig. 5 provides a visual example, where one of the wisdom tooth is miss-detected in the 1st row, while our method can precisely detect and segment any tooth, as depicted in the 2nd row. Moreover, the quantitative result is shown in Table 1. Employing an efficient voting strategy, our approach outperforms HM_{reg} across all four segmentation accuracy metrics, which is consistent with the visual comparison.

4.3.2. Tooth segmentation

Our tooth segmentation component contains three decoder branches, i.e., decoder1 branch, decoder2 branch, and hybrid feature fusion (HFF) branch, respectively. To validate the effectiveness of HFF, we designed two alternative experimental settings for the segmentation module:

1. V1: Utilize decoder1 without skip link, supervising the corresponding segmentation,
2. V2: Employ decoder2 with a skip link, supervising the corresponding segmentation.

The statistical results are listed in Table 2, where our final result achieves the best performance in terms of all segmentation metrics. Specifically, V2 with skip connections performs better than V1, the baseline segmentation configuration. In addition, our result further gains 5.13% and 1.32% Dice score improvements compared to the two decoder versions, attributed to the attention-based HFF scheme's efficient fusion of features from the V1 and V2. Fig. 3 illustrates that, if the CBCT images have blurry tooth boundaries, especially metal artifacts, V2 fails to accurately segment teeth as it heavily relies

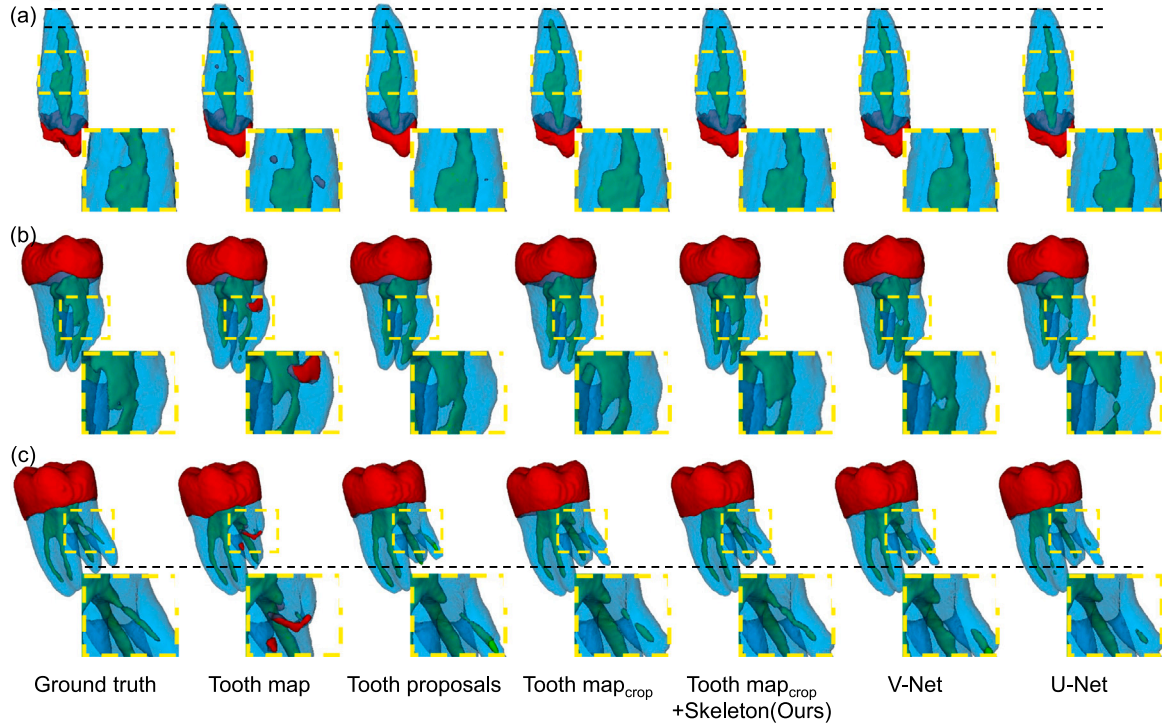
on low-level features. Although V1 utilizes more global features to generate segmentation, the important boundary details are missing in the encoder-decoder architecture, highlighted by the arrows in Fig. 3. In our framework, the supervision of both decoder branches ensures the segmentation encoder does not solely rely on either low or high-level features for tooth segmentation. Additionally, the attention-based fusion scheme efficiently integrates the features from both decoders, leading to robust segmentation results in areas with metal artifacts and limited intensity contrast.

We further compare our framework with three state-of-the-art approaches, including the semantic-based method (MWT-Net [19]), the center-guided network (CGD-Net [20]), and geometry-guided network (HMG-Net [21,22]), as shown in Table 3. Our framework alone detects the tooth centroid with a small localization error of $0.71 \pm 0.46\text{mm}$, highlighting the effectiveness of our tooth detection. Compared with MWT-Net, our method achieves remarkable improvements, demonstrating the advantage of the detect-then-segment framework. Our approach also consistently outperforms CGD-Net, validating the superiority of centroid voting detection and HFF. Lastly, our method slightly surpasses HMG-Net, achieving state-of-the-art performance in this specific task. Fig. 6 provides a quantitative comparison of two typical examples, further emphasizing our method's advantage. Fig. 6(a) illustrates a case with metal artifacts, while Fig. 6(b) reveals an instance where the area of touching teeth exhibits limited intensity contrast. It is worth noting that the segmentation results generated by our approach align more closely with the ground truth. MWT-Net employs tooth contours to divide the foreground tooth label into individual tooth labels, resulting in failures when the common boundaries of neighboring teeth are indistinct, as emphasized by the red boxes. Conversely, CGD-Net, by representing each tooth with a center point, incurs losses of tooth details and struggles to robustly delineate tooth boundaries, particularly in the tooth root area. Additionally, although HMG-Net leverages hierarchical morphological information to guide segmentation and accurately ascertain the number of teeth, it falls short of adequately segmenting tooth morphology in images with artifacts, as highlighted by the red arrows.

Table 3

Statistical comparison with the state-of-the-art methods in terms of tooth segmentation. ‘-’ indicates the metric is inapplicable.

Methods	Localization (mm)	Dice (%)	Precision (%)	95% HD (mm)	ASD (mm)
MWT-Net [19]	-	85.67 ± 0.23	88.93 ± 0.13	1.280 ± 3.01	0.231 ± 0.51
CGD-Net [20]	-	89.14 ± 0.24	92.10 ± 0.13	0.676 ± 1.72	0.156 ± 0.67
HMG-Net [21,22]	-	95.10 ± 0.07	97.86 ± 0.05	0.432 ± 0.76	0.080 ± 0.27
HFF-Net (Ours)	0.71 ± 0.46	95.78 ± 0.06	96.86 ± 0.07	0.449 ± 0.86	0.075 ± 0.29

**Fig. 7.** Visual comparison of different methods by three typical cases. Our method achieves precise segmentation of enamel and dentin, avoiding excessive segmentation, fracture, and short root canals in pulp segmentation. The dashed lines are for the convenience of comparing the lengths of dental pulp and dentin in different methods.**Table 4**

Statistical results of our ablation study by comparing with alternative input in terms of the tooth substructure segmentation.

Methods	Enamel			Pulp			Dentin		
	Dice (%)	95% HD (mm)	ASD (mm)	Dice (%)	95% HD (mm)	ASD (mm)	Dice (%)	95% HD (mm)	ASD (mm)
<i>Tooth map</i>	89.88 ± 0.05	1.697 ± 2.70	0.184 ± 1.17	82.89 ± 0.08	0.723 ± 1.08	0.126 ± 0.85	92.34 ± 0.03	0.590 ± 1.27	0.106 ± 0.75
<i>Tooth proposals</i>	92.20 ± 0.05	0.419 ± 0.99	0.068 ± 0.46	86.55 ± 0.09	0.465 ± 0.65	0.082 ± 0.65	93.96 ± 0.03	0.381 ± 0.84	0.071 ± 0.43
<i>Tooth map_crop</i>	92.32 ± 0.05	0.402 ± 1.23	0.071 ± 0.85	87.01 ± 0.09	0.473 ± 0.64	0.080 ± 0.65	94.06 ± 0.03	0.317 ± 0.36	0.063 ± 0.30
<i>Tooth map_crop + Skeleton (Ours)</i>	92.85 ± 0.04	0.297 ± 0.22	0.051 ± 0.19	87.49 ± 0.09	0.454 ± 0.64	0.076 ± 0.62	94.23 ± 0.03	0.311 ± 0.36	0.060 ± 0.30

Table 5

Statistical comparison of our method with V-Net and U-Net in terms of the tooth substructure segmentation.

Methods	Enamel			Pulp			Dentin		
	Dice (%)	95% HD (mm)	ASD (mm)	Dice (%)	95% HD (mm)	ASD (mm)	Dice (%)	95% HD (mm)	ASD (mm)
V-Net	92.35 ± 0.05	0.364 ± 0.83	0.062 ± 0.62	86.84 ± 0.09	0.454 ± 0.60	0.079 ± 0.53	93.96 ± 0.03	0.306 ± 0.35	0.063 ± 0.29
U-Net	92.76 ± 0.04	0.308 ± 0.39	0.053 ± 0.23	85.16 ± 0.11	0.535 ± 0.68	0.088 ± 0.53	94.10 ± 0.03	0.322 ± 0.36	0.063 ± 0.29
Ours	92.85 ± 0.04	0.297 ± 0.22	0.051 ± 0.19	87.49 ± 0.09	0.454 ± 0.64	0.076 ± 0.62	94.23 ± 0.03	0.311 ± 0.36	0.060 ± 0.30

4.3.3. Tooth substructure segmentation

Owing to the promising results in tooth segmentation, we utilize the same backbone network to expand the segmentation to include tooth substructures. We initially conduct ablative experiments using various inputs to validate the effectiveness of the tooth detection module, tooth segmentation module, and tooth skeleton in substructure segmentation. Here, the *Tooth map* and *Tooth proposals* represent the results of inputting the tooth map and tooth proposals, respectively. The *Tooth map_crop* signifies the results of inputting the product of the tooth map and the tooth segmentation outcome (tooth label). The *Tooth map_crop + Skeleton (Ours)* indicates concatenating tooth skeleton with

Tooth map_crop as the input of tooth substructure segmentation network. Furthermore, we perform comparisons between our approach and commonly well-known networks in medical image segmentation networks (such as V-Net and U-Net), maintaining consistent inputs (i.e., tooth map, tooth label, and tooth skeleton) throughout the comparisons.

The statistical and visual segmentation results are delineated in Table 4, Table 5, and Fig. 7, revealing our proposed method's superior segmentation outcomes. From them, we have made three significant observations. Firstly, the utilization of both the detection module and the tooth segmentation outcomes as inputs yields better substructure

segmentation results compared to using only the outcomes of the detection module. This augmentation is reflected in the metrics reported in the initial three rows of Table 4. Meanwhile, an examination of Fig. 7 elucidates further insights: the *Tooth map* tends to exhibit noticeable over-segmentation of enamel and under-segmentation of dentin and pulp; However, upon incorporating the guidance, i.e., tooth centroid, errors in enamel segmentation are reduced, and improvements are observed in the shape of certain dentin regions near the root apex; By multiplying the tooth label by the tooth map as the input, the network can focus more precisely on foreground substructure segmentation, enhancing the accuracy of dentin shapes and length and pulp details. Secondly, our technique, guided by the tooth skeleton, enhances the segmentation results in statistical and visual. Notably, our method excels over *Tooth map_{crop}*, particularly in enamel and pulp segmentation, leading to about 0.5% improvements in the Dice metric. Meanwhile, the utilization of the tooth skeleton for topological guidance augments pulp length (Fig. 7(a)) and connectivity (Fig. 7(c)), achieving precise and continuity tooth substructure segmentation. Thirdly, the backbone of our method surpasses the V-Net and U-Net. As shown in Table 5, our method performs better in quantitative results, especially in pulp segmentation. From the last three columns in Fig. 7, it can be observed that achieves notably accurate segmentation outcomes for relatively diminutive entities such as dental pulp, specifically in terms of thickness, bifurcation patterns, and length. This achievement can be attributed to the strategic utilization of our distinctive backbone architecture, which effectively capitalizes on an enriched array of pertinent feature information.

5. Discussion

Our methodology achieves automated and accurate segmentation of dental structures, encompassing teeth and their substructures, such as enamel, pulp, and dentin. The procedural framework is divided into three primary stages: tooth detection, tooth segmentation, and substructure delineation. The effectiveness and superiority of our approach are validated through rigorous experimentation.

Commencing with tooth detection, our approach employs centroid clustering to precisely ascertain the spatial coordinates of individual teeth. In contrast to directly regressing centroid coordinates from heatmaps, this strategy attains superior localization accuracy, mitigating the potential oversight of teeth and consequently enhancing the precision of subsequent tooth segmentation. Notably, our tooth segmentation technique integrates a hybrid feature mechanism, harmonizing both low-level and high-level features to effectively preserve tooth contour and boundary information. Particularly noteworthy is the capability of this feature integration to enhance segmentation outcomes for CBCT images afflicted by metal artifacts, thereby contributing to improved results.

In the context of tooth substructure segmentation, the current scientific landscape exhibits a paucity of comprehensive investigations. Thus, our study introduces a method that advances beyond conventional segmentation by further subdividing teeth into finer substructures. Leveraging the inherent skeletal structure of teeth, our approach serves as a guiding framework for pulp segmentation, effectively ameliorating challenges related to pulp discontinuity and inadequate representation. Simultaneously, our approach yields enhanced segmentation metrics for both enamel and dentin delineation.

Although our work has achieved overall promising tooth and substructure segmentations, it still has some limitations in tooth labeling and substructure segmentation of teeth with fillings. First, we have given different labels for teeth based on their centroids, but this is inconsistent with the FDI World Dental Federation notation system [27] as used in clinical practice. Therefore, we will further classify teeth and improve tooth segmentations. Second, our tooth substructure segmentation focuses on healthy teeth, capturing the enamel, pulp, and dentin, but ignores the filling teeth under dental treatment, which is important for treatment and prognosis. In our future work, we will incorporate those treated teeth into substructure segmentation, which will make our method more suitable for digital dentistry.

6. Conclusion

We propose a novel solution for the automatic segmentation solution of tooth and finer substructure from CBCT images, which hinges on robust tooth detection, segmentation, and tooth substructure segmentation. Both qualitative and quantitative evaluations affirm our algorithm's supremacy against state-of-the-art methods, underscoring its remarkable performance and potential clinical applicability for diagnosis and surgical treatment.

CRedit authorship contribution statement

Minhui Tan: Methodology, Experiment, Writing – original draft. **Zhiming Cui:** Supervision, Methodology, Writing – original draft. **Tao Zhong:** Writing – original draft. **Yu Fang:** Writing – original draft. **Yu Zhang:** Supervision, Methodology, Writing – original draft. **Dinggang Shen:** Supervision, Methodology, Writing – original draft.

Declaration of competing interest

None Declared.

Acknowledgment

This work was supported in part by National Natural Science Foundation of China (grant numbers 6230012077, 61971213, 62131015, 62250710165, U23A20295).

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